

Program 10 – Kubernetes Dashboard

STEP 1: Start Minikube with CPU & Memory

minikube start --cpus=2 --memory=4096

STEP 2: Enable & Open Kubernetes Dashboard

minikube dashboard

```
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps$ cd program10
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🔗 Using the docker driver based on existing profile
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
📦 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
  ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, metrics-server, dashboard, default-storageclass
🏠 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
irv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ minikube dashboard
🔍 Verifying dashboard health ...
🚀 Launching proxy ...
🔍 Verifying proxy health ...
🌐 Opening http://127.0.0.1:40291/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
ln: failed to create symbolic link '/home/irv24mc120_yashwanth_c/snap/firefox/7559/.config/dconf/user': File exists
Gtk-Message: 12:06:38.312: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

STEP 3: Create Deployment File (deployment.yaml)

nano deployment.yaml

```
GNU nano 7.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

STEP 4: Create Service File (service.yaml)

nano service.yaml

```
GNU nano 7.2
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

STEP 5: Apply Kubernetes Resources

```
kubectl apply -f deployment.yaml kubectl apply -f service.yaml
```

STEP 6: Monitor Pods (CLI)

```
kubectl get pods
```

STEP 7: Check Nodes

```
kubectl get nodes
```

STEP 8: Check Services

```
kubectl get svc
```

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ nano deployment.yaml
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ nano service.yaml
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ kubectl apply -f service.yaml
service/nginx-service created
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6qrv2   1/1     Running   0           29s
nginx-deployment-6f9664446b-9ckzd   1/1     Running   0           29s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  40d   v1.34.0
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program10$ kubectl get svc
NAME            TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP     10.96.0.1    <none>        443/TCP          5m44s
nginx-service   NodePort      10.111.0.157 <none>        80:32366/TCP     38s
```

STEP 9: Check Pod Logs

```
kubectl logs <pod-name>
```

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program10$ kubectl logs nginx-deployment-6f9664446b-6qrv2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/05 06:38:15 [notice] 1#1: using the "epoll" event method
2026/01/05 06:38:15 [notice] 1#1: nginx/1.29.4
2026/01/05 06:38:15 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/05 06:38:15 [notice] 1#1: OS: Linux 6.14.0-33-generic
2026/01/05 06:38:15 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/05 06:38:15 [notice] 1#1: start worker processes
2026/01/05 06:38:15 [notice] 1#1: start worker process 29
2026/01/05 06:38:15 [notice] 1#1: start worker process 30
2026/01/05 06:38:15 [notice] 1#1: start worker process 31
2026/01/05 06:38:15 [notice] 1#1: start worker process 32
2026/01/05 06:38:15 [notice] 1#1: start worker process 33
2026/01/05 06:38:15 [notice] 1#1: start worker process 34
2026/01/05 06:38:15 [notice] 1#1: start worker process 35
2026/01/05 06:38:15 [notice] 1#1: start worker process 36
2026/01/05 06:38:15 [notice] 1#1: start worker process 37
2026/01/05 06:38:15 [notice] 1#1: start worker process 38
2026/01/05 06:38:15 [notice] 1#1: start worker process 39
2026/01/05 06:38:15 [notice] 1#1: start worker process 40

```

STEP 10: Describe Pod

kubectl describe pod <pod-name>

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/dev0ps/program10$ kubectl describe pod nginx-deployment-6f9664446b-6qrv2
Name:          nginx-deployment-6f9664446b-6qrv2
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Mon, 05 Jan 2026 12:08:04 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.167
IPs:           IP: 10.244.0.167
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://e72799f2d7fa338e02be270c03bdace50581ef0bc17d83c9a92273ad7b94e759
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:ca871a86d45a3ec6864dc45f014b11fe626145569ef0e74deaffc95a3b15b430
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Mon, 05 Jan 2026 12:08:14 +0530
    Ready:         True
    Restart Count:  0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8554b (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized        True
  Ready              True
  ContainersReady    True
  PodScheduled       True
Volumes:
  kube-api-access-8554b:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:          kube-root-ca.crt
    Optional:               false
    DownwardAPI:            true

```

STEP 11: Access Application in Browser

minikube service nginx-service

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

STEP 12: Monitor via Dashboard

